# AI Data Insight Platform Report

Online Sales Data.csv

Generated on March 15, 2026 03:14 PM

| | |
|---|---|
| Rows | 240 |
| Columns | 9 |
| Missing values handled | 0 |
| Duplicate rows removed | 0 |
| Anomalies detected | 28 |

This report includes dataset structure, statistical analysis, charts, explanations, AI insights, anomaly explanations, and correlation analysis.

# 1. Dataset Overview

This section describes the uploaded data after cleaning and preparation.

| | |
|---|---|
| Dataset name | Online Sales Data.csv |
| Rows | 240 |
| Columns | 9 |
| Numeric columns | Transaction ID, Units Sold, Unit Price, Total Revenue |
| Categorical columns | Product Category, Product Name, Region, Payment Method |
| Date columns | Date |
| Anomaly count | 28 |

### Business Context of the Dataset

This dataset can be interpreted as structured transactional sales data that helps explain how customers buy products, how revenue is distributed, and where commercial performance is strongest.

- Customer purchasing patterns show which customers, channels, or segments contribute most to sales activity.
- Product category performance highlights which product groups generate the strongest revenue and demand.
- Pricing influence on revenue helps teams understand how unit price and sales values work together.
- Regional sales performance helps compare branches, cities, or business areas when location columns are available.

Analyzing this kind of dataset helps businesses improve inventory planning, refine pricing strategy, and make better marketing decisions based on real transaction behavior.

# 2. Statistical Analysis

The table below shows the main numeric statistics in simple form.

| Column | Mean | Median | Mode | Min | Max | Std Dev |
|---|---|---|---|---|---|---|
| Transaction ID | 10120.5 | 10120.5 | 10001.0 | 10001.0 | 10240.0 | 69.2814 |
| Units Sold | 2.1583 | 2.0 | 1.0 | 1.0 | 10.0 | 1.3197 |
| Unit Price | 236.3956 | 89.99 | 49.99 | 6.5 | 3899.99 | 428.5511 |
| Total Revenue | 335.6994 | 179.97 | 299.99 | 6.5 | 3899.99 | 484.7913 |

### What these values mean

- Mean: The mean shows the average value of the data.
- Median: The median is the middle value when the values are sorted.
- Mode: The mode is the value that appears most often.
- Minimum: The minimum is the smallest value in the data.
- Maximum: The maximum is the largest value in the data.
- Standard deviation: Standard deviation shows how spread out the values are.

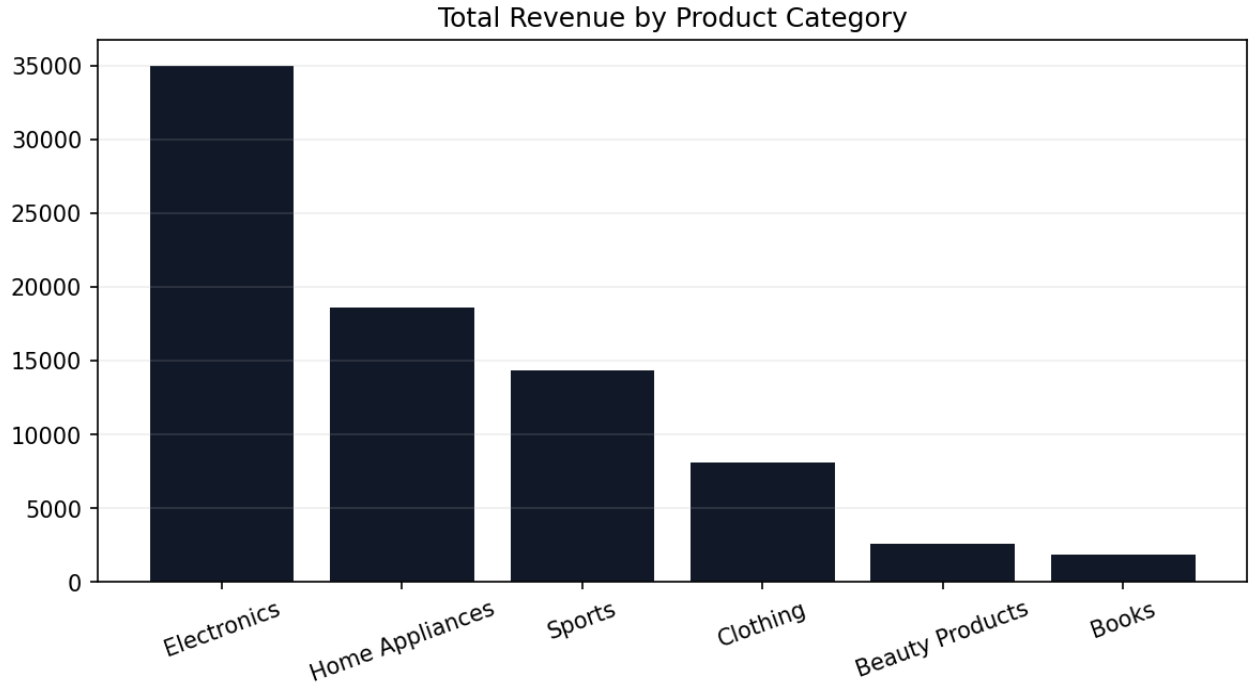
### Business Interpretation of Statistics

- Mean represents the typical value in the dataset and helps estimate normal business performance.
- Median helps describe central tendency without being overly influenced by extreme values or outlier transactions.
- Standard deviation indicates how much pricing or sales values vary, which helps identify stable versus volatile performance.
- Higher variation in pricing may indicate different product segments such as budget products and premium products.
- Businesses can use these statistics to understand pricing distribution, customer demand patterns, and the spread of transaction values.
- The highest observed variability appears in Total Revenue, which suggests this metric should be monitored closely for planning and forecasting.

# 3. Charts

These charts were created automatically to show the most useful patterns in the dataset.

### Total Revenue by Product Category

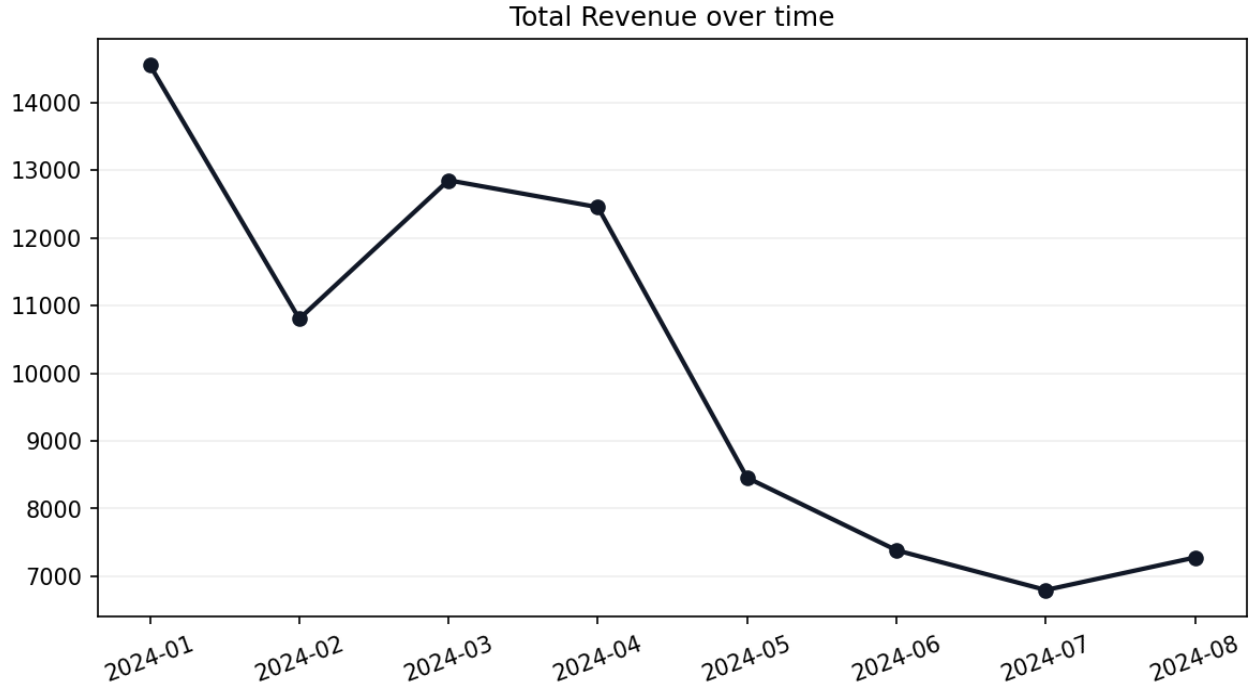



### Chart Interpretation

This chart shows the distribution of values across categories. The tallest bar is Electronics, which has the highest value.

Electronics generates the highest Total Revenue, which suggests that product category deserves close attention for inventory planning, sales focus, and marketing investment.

### Total Revenue over time

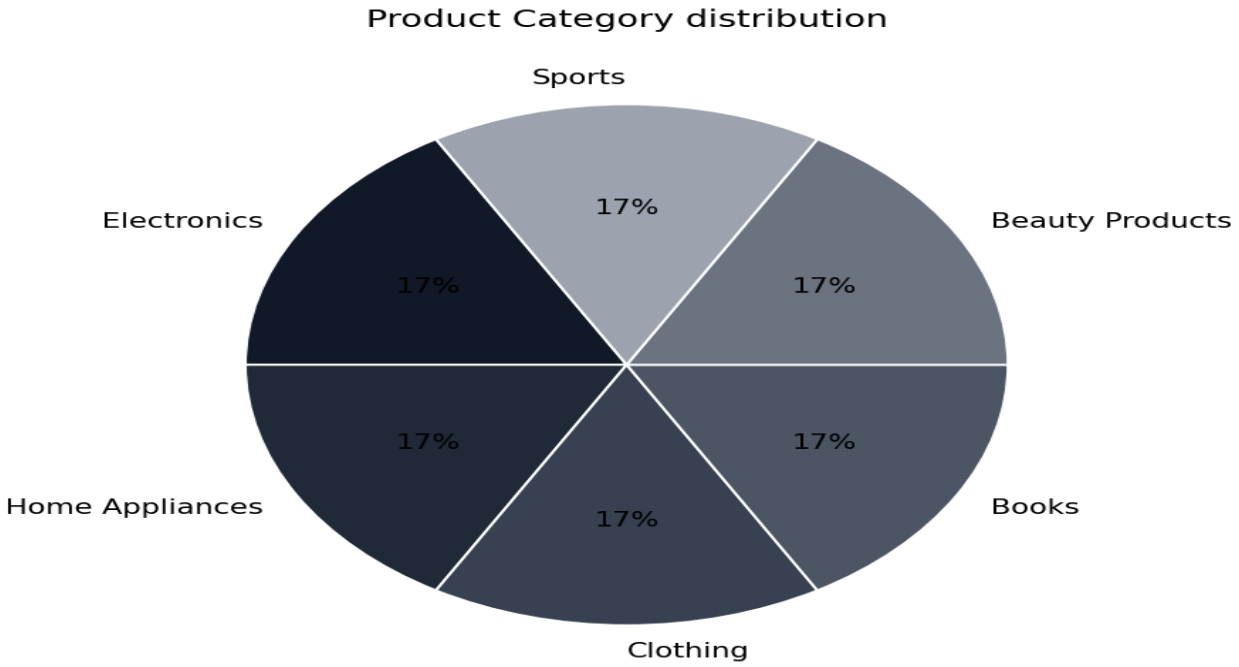


### Chart Interpretation

This chart shows how values change over time. It has a downward trend, which suggests the value is decreasing over time.

The Total Revenue trend is decreasing, which may point to seasonality, weaker demand, or the need for corrective action. Businesses can use this pattern to plan stock, promotions, and timing of campaigns.

### Product Category distribution




### Chart Interpretation

This chart shows the percentage contribution of each category. The largest slice belongs to Electronics.

Electronics represents the largest share of product category at about 16.7%. This helps businesses decide whether to strengthen a leading segment or diversify support across smaller categories.

# 4. Data Explanations

The system cleaned the dataset, calculated statistics, created simple charts, checked for anomalies, and prepared explanations to help you understand the data.

### Data Processing Details

- Duplicate rows removed: 0.
- Missing values handled: 0.
- Descriptive statistics were calculated for numeric columns.
- A small set of charts was generated for quick understanding.
- Anomaly detection checked for unusual values outside the normal range.
- Missing value handling reduces the risk of blank cells distorting averages, totals, and trend calculations.
- Duplicate removal prevents the same transaction or record from being counted more than once in charts and statistics.
- Dataset cleaning standardizes data types and structure so values are easier to compare consistently across the report.
- Anomaly detection looks for unusual values that may represent exceptional transactions, errors, or rare business events.

Together, these preprocessing steps improve the reliability and accuracy of the analysis so business users can trust the reported findings.

# 5. AI Insights

This dataset has 240 rows and 9 columns. The main numeric column is Total Revenue, with an average value of 335.70.

The following insights translate the raw numbers into practical business meaning so non-technical users can quickly understand where attention is needed.

### Highlights

These highlight the most important performance patterns discovered in the dataset.

- The main numeric column is Total Revenue, with an average value of 335.70.
- The top category in Product Category is Electronics based on total Total Revenue.

### Trend Analysis

These observations explain how values change over time and why the direction matters to planning.

- Total Revenue decreases over time in the available date data.

### Relationship Insights

These observations explain how variables move together and how one measure may influence another.

- The strongest correlation is between Unit Price and Total Revenue, and it is positive (0.93).

# 6. Anomaly Explanations

The system detected 28 anomalous row(s). The explanations below describe why these values stand out.

| Row | Column | Value |
|---|---|---|
| 2 | Total Revenue | 1,999.98 |
| 2 | Unit Price | 999.99 |
| 8 | Total Revenue | 2,499.99 |
| 8 | Unit Price | 2,499.99 |
| 9 | Total Revenue | 1,199.98 |
| 9 | Unit Price | 599.99 |
| 14 | Total Revenue | 1,499.98 |
| 14 | Unit Price | 749.99 |
| 20 | Total Revenue | 999.98 |
| 37 | Total Revenue | 1,895 |

### Plain-language explanations

- Row 2: The value 1,999.98 in column Total Revenue is unusually high compared to the normal dataset range of -441.42 to 903.62.
- Row 2: The value 999.99 in column Unit Price is unusually high compared to the normal dataset range of -301.24 to 580.73.
- Row 8: The value 2,499.99 in column Total Revenue is unusually high compared to the normal dataset range of -441.42 to 903.62.
- Row 8: The value 2,499.99 in column Unit Price is unusually high compared to the normal dataset range of -301.24 to 580.73.
- Row 9: The value 1,199.98 in column Total Revenue is unusually high compared to the normal dataset range of -441.42 to 903.62.
- Row 9: The value 599.99 in column Unit Price is unusually high compared to the normal dataset range of -301.24 to 580.73.
- Row 14: The value 1,499.98 in column Total Revenue is unusually high compared to the normal dataset range of -441.42 to 903.62.
- Row 14: The value 749.99 in column Unit Price is unusually high compared to the normal dataset range of -301.24 to 580.73.
- Row 20: The value 999.98 in column Total Revenue is unusually high compared to the normal dataset range of -441.42 to 903.62.
- Row 37: The value 1,895 in column Total Revenue is unusually high compared to the normal dataset range of -441.42 to 903.62.
- Row 37: The value 1,895 in column Unit Price is unusually high compared to the normal dataset range of -301.24 to 580.73.

- Row 38: The value 1,199.97 in column Total Revenue is unusually high compared to the normal dataset range of -441.42 to 903.62.

Only the first 12 anomaly explanations are shown in this report. 32 additional explanations were omitted for brevity.

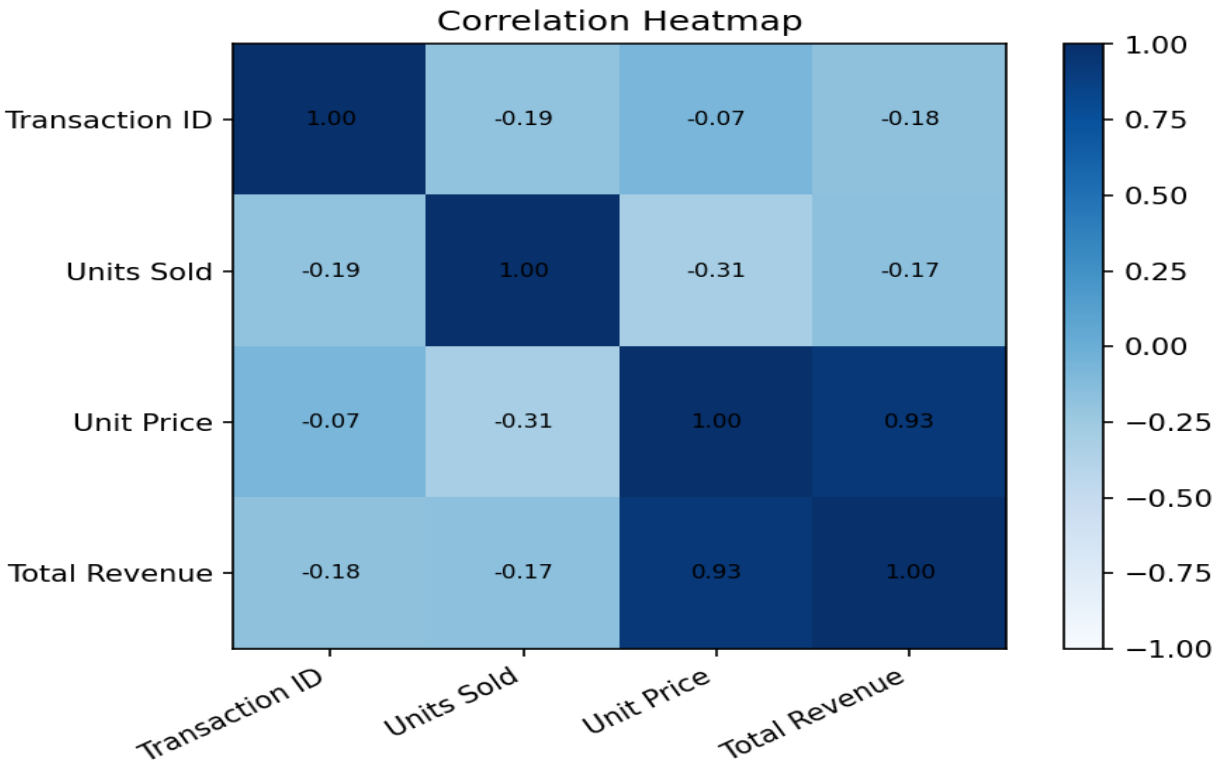
### Business Importance of Anomaly Detection

- Anomalies may represent unusually large transactions that deserve closer operational review.
- They can also indicate potential pricing errors, data entry problems, or unusual customer behavior.
- In some businesses, anomaly detection helps flag possible fraud or activity that falls outside expected norms.
- Monitoring anomalies supports stronger controls, faster investigation, and better data quality over time.

# 7. Correlation Analysis

| Column 1 | Column 2 | Correlation |
|---|---|---|
| Unit Price | Total Revenue | 0.93 |
| Units Sold | Unit Price | -0.31 |
| Transaction ID | Units Sold | -0.19 |
| Transaction ID | Total Revenue | -0.18 |
| Units Sold | Total Revenue | -0.17 |
| Transaction ID | Unit Price | -0.07 |

### Correlation Heatmap




### Business Meaning of Correlations

- Strong correlations help businesses understand which variables move together and which measures should be monitored as a group.
- These relationships can support pricing decisions, promotional planning, and performance management.
- The strong correlation between Unit Price and Total Revenue suggests that pricing strategies directly influence revenue performance, so pricing decisions should be reviewed carefully.

# 8. Business Recommendations

The suggestions below translate the analysis into practical actions that business teams can use for planning, monitoring, and growth.

- Focus marketing, inventory, and promotional effort on Electronics, because it is currently the highest-performing category by total Total Revenue.
- Investigate seasonal patterns, pricing decisions, and customer demand drivers to understand the declining trend and respond early.
- Analyze seasonal sales patterns over time to improve timing for campaigns, discounts, and stock replenishment.
- Investigate anomaly transactions to confirm whether they represent legitimate high-value sales, pricing issues, or unusual activity that needs control review.
- Monitor strongly related variables together so pricing models and promotional strategies are based on how business measures influence one another.
- Use forecasting for future planning so teams can prepare inventory, staffing, and marketing activity before demand changes occur.

# 9. Future Enhancements

Future versions of the AI Data Insight Platform can expand from descriptive reporting into more advanced decision support and predictive analytics.

- Predictive sales forecasting could estimate future demand and support more accurate planning.
- Customer segmentation analysis could group buyers by behavior, value, or purchasing preferences.
- Real-time monitoring dashboards could alert teams quickly when performance changes or anomalies appear.
- Advanced machine learning models for demand prediction could improve replenishment, pricing, and promotional decisions.

These improvements would help businesses move further toward data-driven decision making and faster operational response.